Modern Artificial Intelligence

Advanced Paradigms in AI (2026)

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Knowledge Representation

Problem-Solving

State-Space Representation

Covering the three foundational pillars that power every modern AI system

Course Overview

01

Knowledge Representation

How AI stores, structures, and reasons with information — vectors, graphs, and symbolic logic.

02

Problem-Solving Approaches

CSPs, Tree of Thoughts, and Agentic AI — how AI finds the best action to take.

03

State-Space Representation

Mathematical maps that let AI navigate environments — from puzzles to drone delivery.

Part 1: Knowledge Representation

How AI understands, stores, and reasons with the world

What is Knowledge Representation (KR)?

Raw Data
Numbers, text, images, signals

Knowledge Representation
Structure, rules, relationships

AI Intelligence
Reason, learn, decide

Data ≠ Intelligence

Raw data alone doesn't create intelligence. AI must transform data into structured, meaningful knowledge.

The Synergy

AI needs both knowledge (facts, rules) AND intelligence (the ability to apply those rules to real situations).

The Goal

Format information so machines can reason, learn, and make decisions — just like humans do.

Two Schools of AI Thought

⚙ Symbolic AI		🧠 Connectionist AI	
Approach	Explicit rules & logic	**Approach**	Neural networks & patterns
Transparency	High — decisions traceable	**Transparency**	Low — often a black box
Strength	Logical reasoning, math	**Strength**	Images, language, predictions
Weakness	Struggles with messy data	**Weakness**	Hard to explain decisions
Example	IF fever > 38°C THEN check infection	**Example**	ResNet identifies tumor in X-ray LLMs Hallucinating

VS

2026 Insight: Advanced AI combines BOTH approaches for maximum accuracy and explainability

Vector Embeddings

The Bridge Between Raw Data and AI Understanding

A **Vector Embedding** converts a complex piece of data — a paragraph, an image, a song — into a list of numbers (a vector). In this numerical space, *things that are similar end up close together*. This is how AI understands meaning, not just keywords.

Simple Analogy

Imagine placing words on a map. "Apple" and "Pear" would be placed close together (both are fruits). "Apple" and "Car" would be far apart. Vector databases store these "maps" and let you find the nearest neighbors instantly.

Real example: The vector for "heart attack" will be geometrically close to "myocardial infarction" — a critical capability for medical AI that needs to match patient records with research papers.

Vector Databases in 2026

Specialized databases store and search these embeddings at scale:

- **Pinecone** — cloud-native, used for enterprise RAG pipelines
- **Milvus** — open-source, deployed on-premise for sensitive data
- **Weaviate** — combines vector search with knowledge graph features

They enable **similarity search** — finding relevant information even when exact keywords don't match.

Vector Embeddings — The Connectionist Approach

Core Concept

Words, images, and sounds are converted into lists of numbers (vectors). Similar concepts are placed close together in mathematical space — allowing AI to do math with language.

1. **Convert to Numbers**

   Each word/image → a list of 768–1536 numbers (vector) capturing meaning.

2. **Map to Space**

   Similar concepts are placed near each other in high-dimensional space.

3. **Reason Geometrically**

   AI performs arithmetic on vectors to infer analogies and relationships.

Powers: ChatGPT, BERT, GPT-4

https://projector.tensorflow.org/

Word Analogy Math

KING

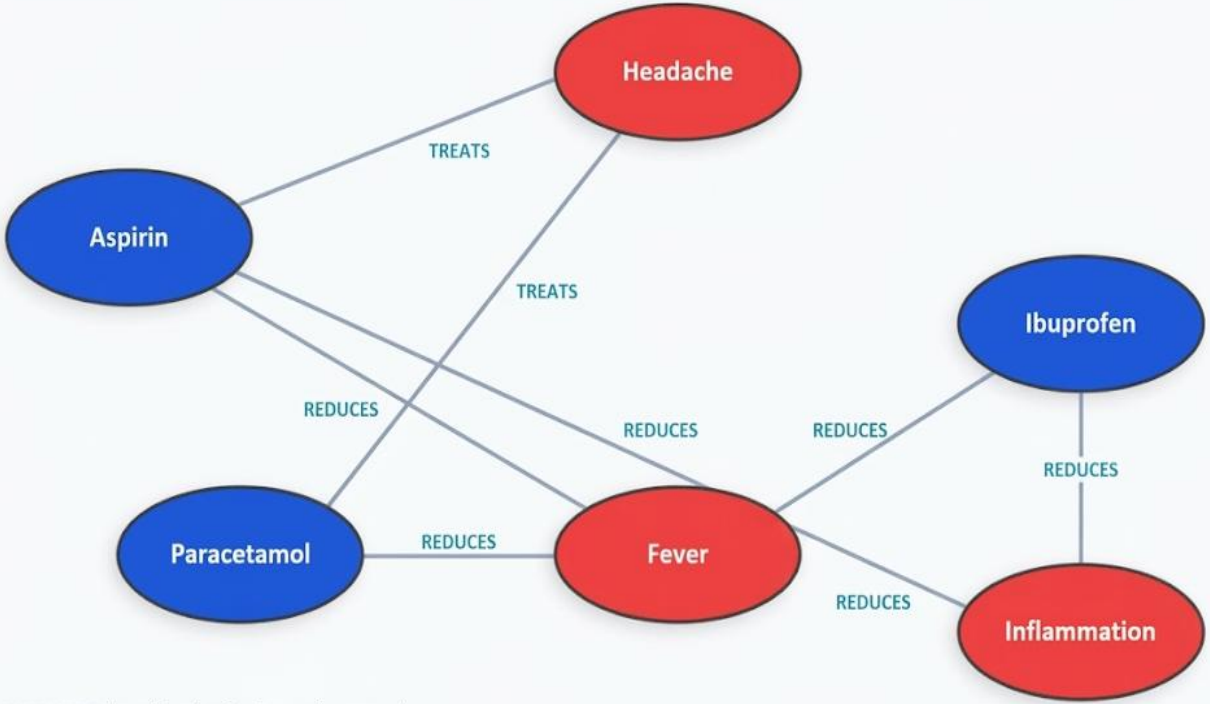
− MAN

+ WOMAN

= QUEEN ✓

AI uses geometry of meaning to compute language!

Knowledge Graphs — The Symbolic Approach



REDUCES (cooling/reducing color tone)
TREATS (restorative/healing tone)

Why KGs Matter

Embeddings show two things are related. Knowledge Graphs explain exactly HOW and WHY — using labeled edges between entities.

KG Structure

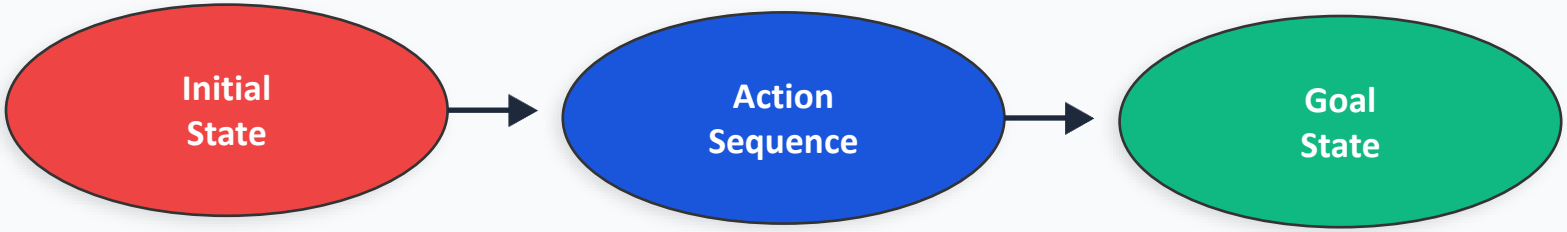
Nodes: Real entities (drugs, diseases)
Edges: Exact relationships (TREATS, CAUSES)
Inference: Multi-hop reasoning

Used by: Google Search, IBM Watson Healthcare, Enterprise Supply Chains

Part 2: Problem-Solving Approaches

CSPs, Tree of Thoughts, and Agentic AI

How AI Navigates from Start to Finish



Constraint Satisfaction (CSPs)

Declare what makes a VALID solution using variables, domains, and constraints. No blind searching.

Tree of Thoughts (ToT)

Explore multiple reasoning paths simultaneously. Backtrack when a branch fails — like a chess player thinking ahead.

Agentic AI (ReAct Loop)

Autonomous agents that Reason → Act → Observe in a loop to complete real-world tasks without human input.

Constraint Satisfaction Problems (CSPs)

1 Variables

What we need to figure out
e.g. Which nurse works which shift?

2 Domains

The possible options available
e.g. Morning, Evening, Night shift

3 Constraints

The strict rules to follow
e.g. No nurse works two-night shifts in a row

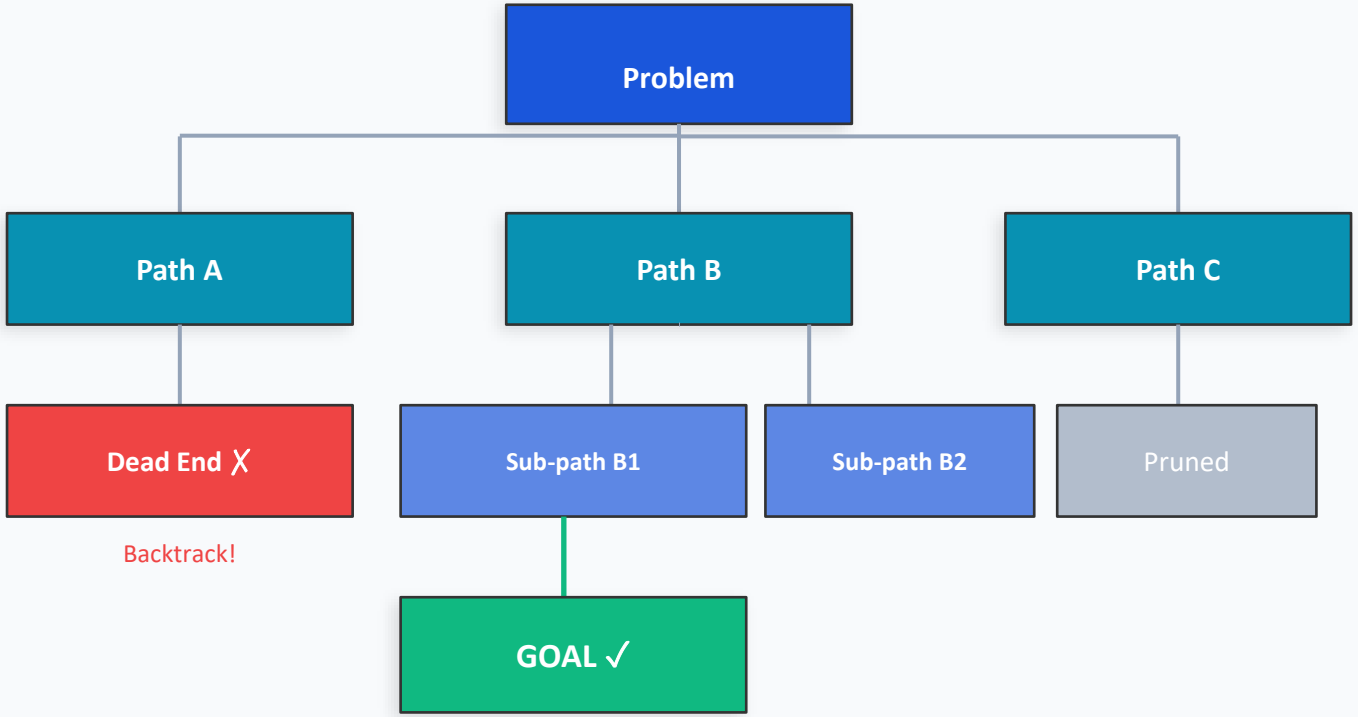
Sudoku Analogy

Variables:	Empty boxes in the puzzle
Domains:	Numbers 1–9
Constraints:	No repeat in row, column, or 3×3 box
Solution:	A valid filled grid — found without guessing!

Real-World Applications

- Semiconductor chip design (routing transistors)
- Airline flight scheduling & crew allocation
- Factory logistics and assembly line timing
- University course and exam timetabling

Tree of Thoughts (ToT) Framework



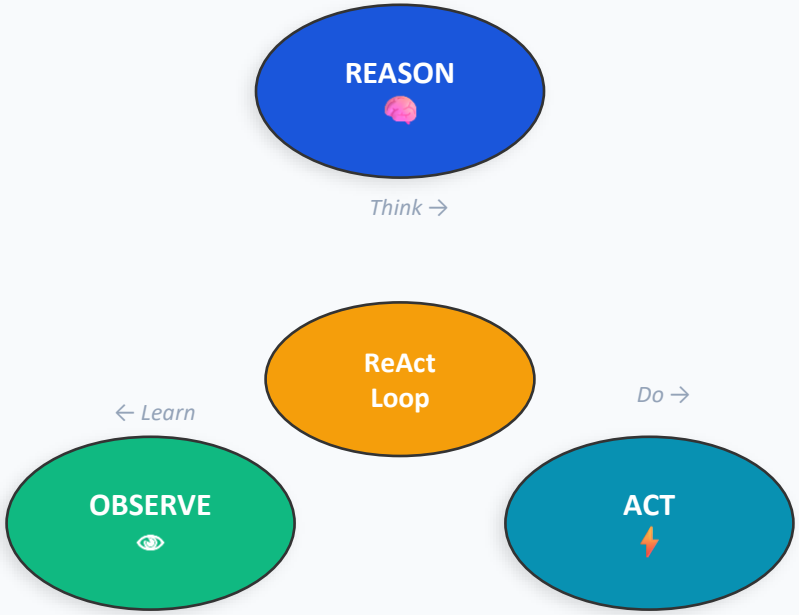
CoT vs ToT

Chain **o**f **T**hought: Linear, one path

Tree **o**f **T**houghts:

Multiple parallel paths, backtracking, evaluation

Agentic AI — From Assistants to Autonomous Workers



Memory

Stores context across steps

Tool Use

Browses web, runs code, clicks UI

Collaboration

Works with other AI agents

Goal-Driven

Acts until objective is achieved

Example: Healthcare billing agent detects a rejected claim → checks patient records → finds missing code → resubmits automatically — no human needed

Part 3: State-Space Representation

Mathematical maps of every possible configuration

State-Space Representation — Mapping All Possibilities

State

A snapshot of the world at a given moment
e.g. Position of tiles, volume of water in jugs

Actions

Legal moves that transition between states
e.g. Slide tile left, pour water from jug A to B

Initial State

Where the AI starts — the beginning configuration

Goal State

The target — what a valid solution looks like

Initial State → Search via Algorithm → **Goal State**

Key Idea: The AI builds a graph of all reachable states, then finds the shortest/cheapest path from start to goal.

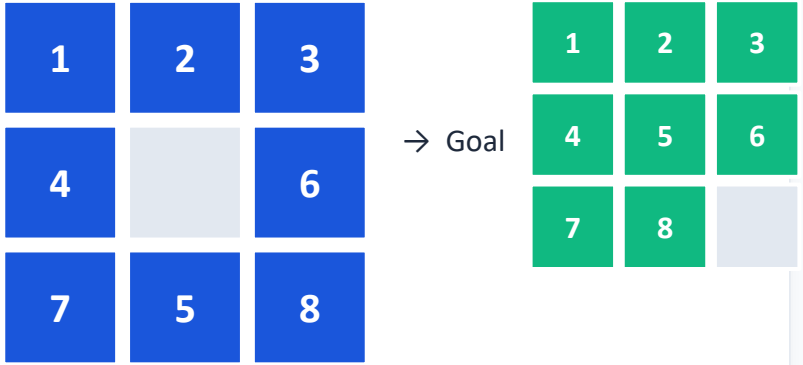
Classic State-Space Examples

Water Jug Problem

State: (x, y) — volume in each jug

`(0, 0)`	Start
`(0, 3)`	Fill 3-gal jug
`(3, 0)`	Pour into 4-gal
`(3, 3)`	Fill into 3 –gal jug
`(4, 2)`	Pour 3-gal jug into 4-gal
`(0, 2)`	Empty 4 Gal
`(2, 0)`	Pour in 4 GAL → ... reached goal

8-Puzzle Problem



State: Matrix arrangement of tiles

Variables:	Position of each tile in the 3×3 grid
Actions:	Slide blank space up, down, left, or right
State Space:	9! = 362,880 possible arrangements
Algorithm:	A* search finds optimal solution quickly

PROBLEM-SOLVING USING STATE-SPACE REPRESENTATION

1: DEFINE STATE REPRESENTATION
Clearly represent the current situation

2: DEFINE ACTIONS (PRODUCTION RULES)
Identify all possible operators

- **PRECONDITIONS** (When action applies)
- **EFFECTS** (What changes after)

3: SPECIFY THE INITIAL STATE
Define the starting point of the problem

4: DEFINE THE GOAL STATE
Clearly specify the goal condition to meet

5: APPLY PRODUCTION RULES (SEARCH PROCESS)
Start from Initial State, apply rules to generate new states, repeat until Goal reached

VISUAL FLOW SUMMARY

State-Space in the Real World (2026)

Drone Delivery Navigation

Drones fly in continuous 3D space with uncertainty from wind and battery. AI uses transformers to compress visual data into compact state spaces, enabling real-time, energy-efficient routing.

Smart Power Grids

The electrical grid is a constantly shifting state space. AI acts as a 'Grand Conductor', shifting EV charging and load balancing in milliseconds based on weather and market prices.

Global Logistics

AI models entire global supply chains as a state space, dynamically rerouting delivery trucks and ships the moment a storm or delay occurs — saving millions in costs.

Medical Diagnosis

Patient health evolves as a state space. AI tracks symptoms, lab results, and treatment responses over time to suggest optimal treatment pathways and flag drug interactions.

Summary — The 3 Pillars of Modern AI

1

Knowledge Representation

Vector Embeddings — semantic math in high-dimensional space | Knowledge Graphs — structured facts with labeled relationships | Hybrid approach combines both for accuracy + explainability

2

Problem-Solving Approaches

CSPs — declare constraints, AI finds valid solutions | Tree of Thoughts — multi-path reasoning with backtracking | Agentic AI — autonomous Reason → Act → Observe loops

3

State-Space Representation

States + Actions define the universe of possibilities | C-Space simplifies robot path planning mathematically | Powers drones, grids, logistics, and medical AI

These three pillars work together — every advanced AI system in 2026 uses all of them.

Thank You

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